IED-GCN: An Internal and External Decoupled Graph Convolutional Network for Landslide Susceptibility Assessment

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***Abstract*—Landslides are one of the most frequent and destructive geological disasters, often causing significant threats to human life and infrastructure. Landslide susceptibility assessment (LSA) plays a vital role in disaster prevention and risk management. Graph convolutional networks (GCNs) have demonstrated strong potential in LSA due to their ability to model graph-structured data. However, existing GCN-based models often suffer from feature aggregation issues, where the neighborhood aggregation of landslide conditioning factors (LCFs) leads to information loss and reduced predictive performance. To address this limitation, we propose an internal and external decoupled GCN (IED-GCN) based on superpixel segmentation. First, the simple linear iterative clustering (SLIC) algorithm is employed to segment the study area into superpixels, creating more meaningful spatial units. Then, the proposed model applies internal graph convolution (IGC) to capture local feature dependencies and external graph convolution (EGC) to model global feature relationships. Furthermore, edge decoupling is incorporated into the EGC to mitigate overfitting and enhance the model's generalization capability. The effectiveness of IDE-GCN is validated using a dataset comprising historical landslide occurrences and 13 LCFs in the Three Gorges Reservoir area. The dataset is divided into 70% of the data for training and 30% for testing. The experimental results demonstrate that our IED-GCN significantly outperforms conventional models, such as convolutional neural networks (CNNs), residual convolutional networks (ResNets), dense convolutional networks (DenseNets), and miniGCN, achieving F1 score, Kappa coefficient, and AUC values of 0.9839, 0.9675, and 0.9838, respectively. In summary, our IED-GCN effectively addresses the challenge of feature aggregation in LSA tasks, offering superior performance in geological disaster prediction with complex spatial data.**

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***Index Terms*—Deep learning (DL), edge decoupling, graph convolutional network (GCN), landslide susceptibility assessment (LSA).**

I. Introduction

LANDSLIDES are cascading geo-disasters that can pose a significant impact on human lives, infrastructure, and the environment, while also serving as a key driver of landscape evolution [1], [2]. According to the *China Statistical Yearbook of Geological Disasters* 2024, approximately 348 140 geological disasters occurred in China between 2000 and 2023, of which more than 246 632 (71%) were landslides. These disasters have resulted in tens of thousands of casualties and caused direct economic losses exceeding tens of billions of dollars. Beyond their immediate impacts, landslides disrupt infrastructure, degrade ecosystems, and hinder socio-economic development. Therefore, accurate landslide susceptibility assessment (LSA) is of great significance for disaster early warning, risk management, and land use planning [3].

LSA refers to the likelihood of landslide occurrences based on local terrain conditions [4]. Existing LSA models can be broadly divided into physics-based models [5], [6], statistical models [7], and machine learning (ML) models [8], [9]. Physics-based models rely on fundamental geotechnical principles and mathematical equations to simulate slope stability conditions [10]. Statistical models, such as the information value method [11], frequency ratio method [12], and logistic regression method [13], establish empirical relationships between past landslides and influencing factors to generate landslide susceptibility maps [14], [15]. In contrast, ML-based approaches leverage powerful nonlinear modeling capabilities to identify complex patterns in large-scale geospatial data. Common ML techniques include support vector machines [16], random forest [17], decision trees [18], and artificial neural networks [19]. However, ML methods have limitations in fully capturing the interactions among environmental factors, are sensitive to data quality, and are prone to getting stuck in local optima, which restricts their overall performance. With the increasing availability of high-resolution remote sensing data and advances in computational capabilities, deep learning (DL) has emerged as a dominant tool for LSA. DL consists of hierarchical layers that automatically extract complex spatial features from multisource geographic and environmental

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TABLE I
RECENT LITERATURES ON THE APPLICATION OF GCN IN LANDSLIDE STUDY

Literature	Year	Study area	LSA unit for GCN	Remark
Wang *et al.*	2023	Wanzhou District in the Three Gorges reservoir area	Slope	The identified potential landslides and historical landslides are jointly used as the dataset for LSA of GCN.
Xia *et al.*	2023	Lueyang County in Shaanxi Province	Slope	The graph structure based on slope elements is generated by using the KNN algorithm for the GCN model.
Zhang *et al.*	2024	Bailong River Basin	Slope	A geographic environment similarity-constrained adjacency graph based on geographic node features is constructed for GCN.
Chen *et al.*	2023	Yanbian Korean Autonomous Prefecture	Pixel	Spatial and channel features are extracted using depth- separable convolution and processed separately with GCN.

datasets, enabling improved landslide prediction accuracy [20]. In recent years, many scholars have applied DL models to LSA. For instance, Nikoobakht et al. [21] demonstrated the superiority of convolutional neural networks (CNNs) over traditional ML models in LSA. However, conventional CNN architectures often suffer from limited spatial feature extraction capability due to insufficient localization information. To solve this problem, Hajimoradlou et al. [22] introduced U-Net into LSA, allowing for enhanced ground surface tracking and contour alignment. Similarly, Ge et al. [23] explored CNN variants, such as ResNet and DenseNet, for landslide susceptibility mapping along transmission lines, achieving over 90% accuracy in both training and testing. In addition, recurrent neural networks (RNNs) have been employed in LSA. Ngo et al. [24] compared CNN and RNN models on a national scale LSA, demonstrating that RNN outperforms CNN by effectively capturing both spatial dependencies and temporal characteristics of landslides.

Although DL has made significant progress in the field of LSA, traditional DL models are primarily designed for Euclidean-structured data and struggle to effectively handle irregular spatial structures. Landslides often exhibit complex boundaries and shapes, making it difficult for traditional DL models to accurately capture their spatial distribution patterns. In contrast, graph-based models offer a more flexible representation of the complex spatial dependencies inherent in landslides. The introduction of graph convolutional networks (GCNs) by Kipf and Welling [25] marked a breakthrough in applying DL techniques to non-Euclidean domains. The principle of GCNs is to generate higher-level node feature representations by aggregating information from neighboring nodes [26]. In recent years, the application of GCNs in LSA has been continuously expanding (see Table I). Wang et al. [27] applied GCNs to slope unit-based geological maps, effectively modeling the spatial topological relationships between different geomorphological units, thereby enhancing the model's ability to identify landslide-prone areas. Similarly, Xia et al. [28] constructed graph structures among slope units using the k-nearest neighbor (KNN) method to strengthen the modeling of spatial associations between land surface elements. Furthermore, Zhang et al. [29] proposed a GCN framework that integrates geographical environment similarity and sample balancing strategies to improve model robustness under imbalanced data scenarios. In addition to slope unit-based graph construction, researchers have also explored pixel-based and superpixel-based strategies to enhance the representation of landslide features at different spatial scales [30]. Chen et al. [31] developed a GCN model based on pixel units, where each raster pixel is treated as a graph node. By combining depth-separable convolution for spatial and channel feature extraction, this approach effectively captures fine-grained spatial variations. In contrast, Wan et al. [32] proposed a superpixel-based GCN method, which segments the image into spatially coherent regions and uses them as nodes in the graph. This allows for better modeling of regional-scale spatial structures and provides new insights into capturing large-scale landslide distribution patterns.

However, the graph structures of these GCNs used for LSA are all constructed at a single unit, making it difficult to simultaneously capture both local details and global structures of landslide distribution. This limitation results in incomplete feature representations and hampers the model's ability to characterize landslide-inducing factors. To overcome these shortcomings, this article introduces an internal and external decoupled GCN (IED-GCN) based on superpixel segmentation. By constructing graph structures at both the pixel scale and the superpixel scale simultaneously, the proposed method enables more comprehensive spatial information modeling. The proposed approach is evaluated on a landslide dataset constructed using historical records and 13 landslide conditioning factors (LCFs) from the Three Gorges Reservoir area. Comparative experiments with four benchmark models validate the effectiveness of our newly developed IED-GCN.

The main contributions of this article can be summarized as follows.

1) We proposed a novel graph-based framework, called IED-GCN, specifically designed for landslides susceptibility assessment. IED-GCN was tailored to preserve the integrity of landslide features and ensure that each computational unit effectively represented the geological structure associated with the landslide.
2) The internal graph convolution (IGC) module captures geological heterogeneity and fine-grained topographic variations, while the external graph convolution (EGC) module simulates the spatial interactions among topo-

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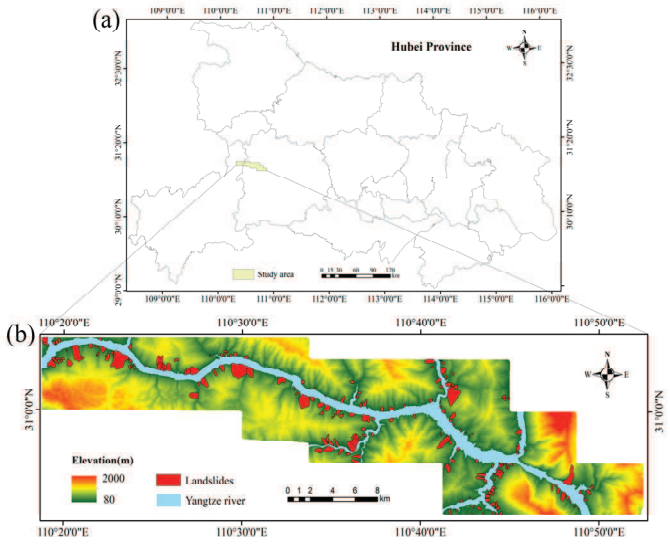


Fig. 1. Location of the study area. (a) Geographical location of the study area. (b) Landslide distribution in the study area.

graphic units, enabling a robust characterization of complex landslide-triggering mechanisms.

3) The edge decoupling module reduces feature redundancy and improves the model's adaptability and discriminative capability under complex terrain conditions.
4) Extensive experiments on the Three Gorges Reservoir area, along with comparisons to four benchmark methods and detailed ablation studies, confirm the effectiveness and robustness of the proposed IED-GCN.

II. Study Area and Data

A. Study Area

The study area is situated in the Three Gorges Reservoir Area of Hubei, China, encompassing the intersection of Zigui, Yichang, China, and Badong, Enshi, China, as shown in Fig. 1(a). Geographically, the region spans from 30°1′12″N to 30°56′58″N in latitude and from 110°18′0″E to 110°52′30″E in longitude, covering approximately 396 km^2. This region is characterized by highly diverse topography, with elevations ranging from 80 to 2000 m above sea level, leading to significant variations in terrain and slope. The geological structure is complex, featuring frequent fault lines, folds, and unstable lithology, making it highly susceptible to landslide occurrences.

The study area lies within a subtropical monsoon climate zone, experiencing warm and humid conditions. The annual average precipitation is approximately 1150 mm, with most of the rainfall occurring between June to September during the rainy season [33]. Frequent heavy rainfall during this period, combined with the region's steep slopes, fractured geological formations, and unstable rock and soil layers, significantly contributes to recurrent landslide events. As illustrated in Fig. 1(b), landslides are predominantly concentrated in high-risk zones, particularly along the reservoir banks and areas near river valleys. Given these conditions, accurate LSA in the Three Gorges Reservoir area is of great significance for infrastructure safety and regional stability.

B. Landslide Inventory Map

DL models require extensive, high-quality ground truth data to ensure stable and reliable results [34]. In the context of LSA, a landslide inventory map serves as a fundamental component for geological disaster research. This map not only visualizes the spatial distribution of historical landslides but also provides essential labeled data for model training and landslide susceptibility analysis.

In this study, a comprehensive landslide inventory map was constructed through the integration of remote sensing image interpretation, field investigation data, and historical landslide records. A total of 202 landslides were identified, covering approximately 23.40 km^2. The spatial distribution of these landslides reveals a clear concentration along the reservoir banks of the Yangtze River, where steep slopes and fluctuating water levels create highly susceptible conditions for landslides. In addition, the complex geological structure and intense human activities in the area further exacerbate landslide susceptibility. This inventory is essential for LSA and contributes to informed risk mitigation strategies in the region.

C. Landslide Conditioning Factors

The occurrence of landslides is a complex process that is influenced by geological conditions and environmental factors [8]. To comprehensively assess landslide susceptibility in the Three Gorges Reservoir area, we selected 13 LCFs based on an extensive review of existing literature and field investigations (see Table II), including topographic, geological, and human activity-related influences [35].

Among them, the topographic factors include elevation, slope, aspect, bedrockslope, terrain position index (TPI), terrain ruggedness index (TRI), stream power index (SPI), and topographic wetness index (TWI), all extracted from digital elevation models (DEMs). These variables are essential for characterizing terrain features, hydrodynamic conditions, and their influence on slope stability. Geological factors, derived from geological maps, consist of lithology and distance to fault (DTF). The former helps assess the stability of bedrock, while the latter indicates potential structural weaknesses. Human activity-related factors include distance to road (DTRO) and distance to river (DTRI), obtained from a road map and a hydrological map, respectively. Road construction may destabilize slopes by altering surface drainage patterns, and river incision or toe erosion can further contribute to slope failure. In addition, the normalized difference vegetation index (NDVI), calculated from Landsat-8 OLI imagery, is used to represent the stabilizing effect of vegetation cover on slopes. By integrating the above 13 LCFs, a solid database was provided for LSA modeling.

III. Methodology

The overall workflow of this study is shown in Fig. 2. The methodology consists of three primary stages: 1) data

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TABLE II
SOURCES OF THE RAW DATA USED FOR LCFS

LCFs	Raw Data	Data Type	Scale	Source
-	Landslide inventory	Vector	1:10 000	remote sensing images, field investigation and historical landslide data
Elevation, Aspect, Slope, Bedrockslope, TPI, TRI, SPI, TWI	DEM	Raster	30m	ALOS (Advanced Land Observing Satellite)
NDVI	Landsat-8 OLI	Raster	30m	Geospatial Data Cloud website, USGS
Lithology, DTF	Geological map	Vector	1:200 000	National Geological Archives
DTRO	Road map	Vector	1:250 000	OSM
DTRI	Hydrological map	Vector	1:250 000	OSM

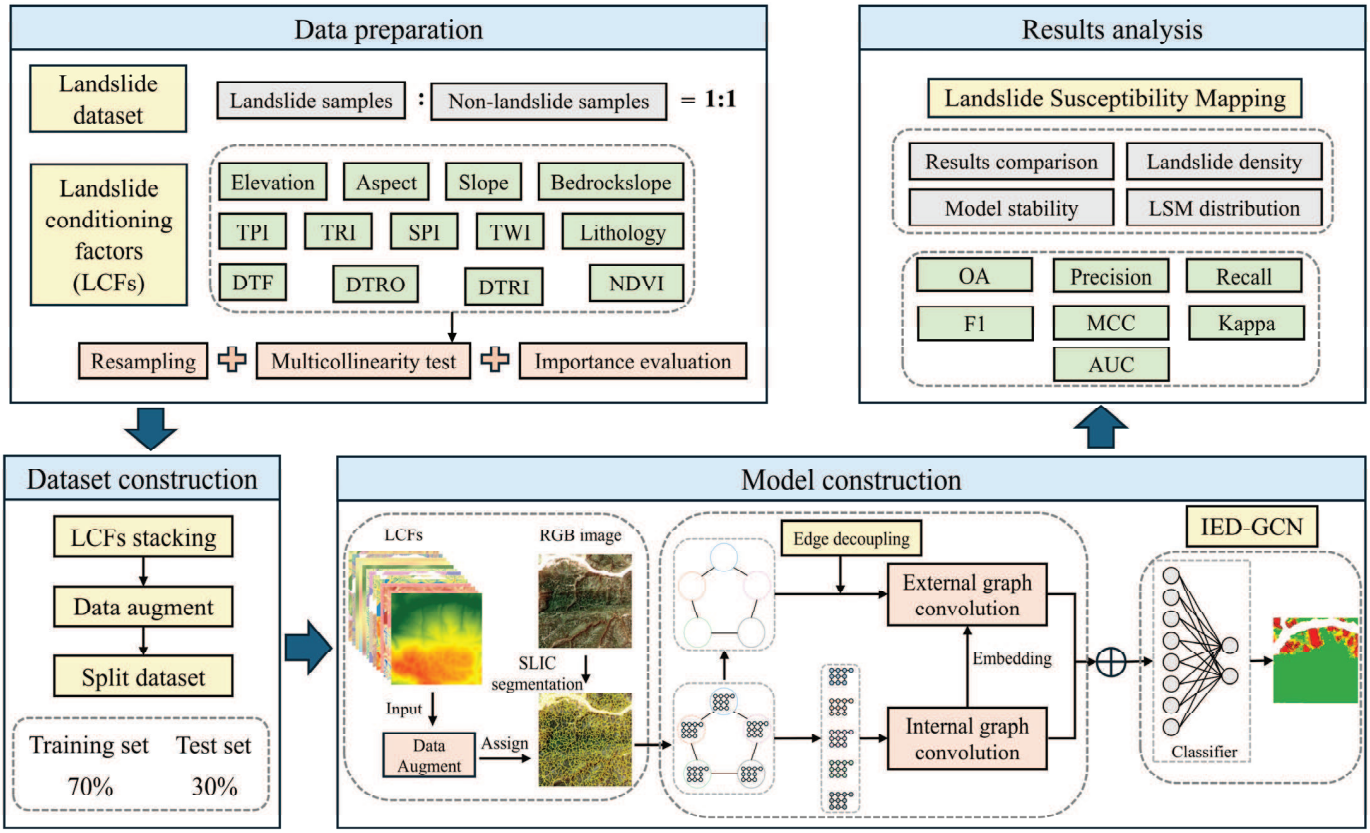


Fig. 2. Flowchart of the conducted study.

preparation, including the compilation of the landslide inventory and LCFs, and partitioning of the dataset into training and testing subset with a 7:3 ratio; 2) construction of IED-GCN model; and 3) performance evaluation and comparative analysis against four representative models, namely CNN [36], ResNet [37], DenseNet [38], and miniGCN [39].

A. Data Preparation

1) Data Preprocessing: A variety of spatial datasets relevant to LSA were collected and harmonized within a unified coordinate system. To ensure comparability and suitability for modeling, all raw data were resampled to a uniform spatial resolution. In this study, we opt for the grid unit with a size of 28.5 × 28.5 m as the evaluation unit, a preference frequently endorsed by researchers in diverse domains for classification or regression tasks [40], [41]. In addition, to mitigate the influence of varying data scales, all feature data underwent normalization, ensuring that each feature value was mapped within a consistent numerical range. This normalization process prevents the excessive dominant influence from specific features during model training and enhances data reliability and consistency.

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2) Selection of LCFs: The selection of LCFs is very important in LSA, as it directly affects the accuracy and reliability of the model [42]. To ensure that selected factors are both independent and representative of landslide occurrence, a multicollinearity test [43] was conducted, followed by importance ranking based on the Gini coefficient [44].

To avoid potential instability and loss of predictive capability due to high correlation among factors, the variance inflation factor (VIF) and tolerance (TOL) were employed in the multicollinearity test. The VIF quantifies the degree of collinearity between an independent variable and other independent variables, and is expressed as

$$\mathrm{VIF} = \frac{1}{1 - R^2} \tag{1}$$

where R^2 represents the coefficient of determination between a given independent variable and all other independent variables. The TOL is defined as the reciprocal of VIF. A VIF value exceeding 10 indicates severe multicollinearity, necessitating further processing or exclusion of the affected factor. By calculating the VIF values for all candidate factors, we systematically eliminated or transformed highly collinear variables, ensuring the independence of factors within the model.

To further refine the selection of LCFs, we applied the Gini index to rank factor importance. The Gini index quantifies the contribution of each feature to classification accuracy by assessing its information gain during decision tree splits. This ranking process allows identification of the most influential factors driving landslide occurrence, providing a robust basis for feature selection.

3) Dataset Construction: Following the multicollinearity test and importance ranking, the final training and testing datasets for the LSA model were constructed through the following steps. First, the selected LCFs were overlaid to create a multidimensional feature dataset. Second, an equal number of nonlandslide and landslide samples were selected to form a balanced dataset for model training. Subsequently, the dataset was randomly partitioned into a training set and a testing set according to a ratio of 7:3. Finally, data augmentation techniques, including noise addition and color jittering, were applied to enhance the diversity of the training set and improve model robustness.

B. Proposed IED-GCN Model

1) GCN Algorithm: A GCN is a neural network architecture designed to aggregate node information from neighboring nodes through convolution operations, leveraging the inherent structure of the graph [45]. The core idea of GCN is to learn node representations through convolution operations applied to the features of nodes and their neighbors, thereby capturing spatial structure information in the graph structure (see Fig. 3).

A graph is formally represented as $G = (V, E)$, where V is the set of nodes, and E is the set of edges. The feature vector of the node v_i is denoted as $h_i \in \mathbb{R}^d$. The goal of the graph convolution operation is to iteratively update the feature representations of each node based on its neighborhood information. The graph convolution operation is defined as follows:

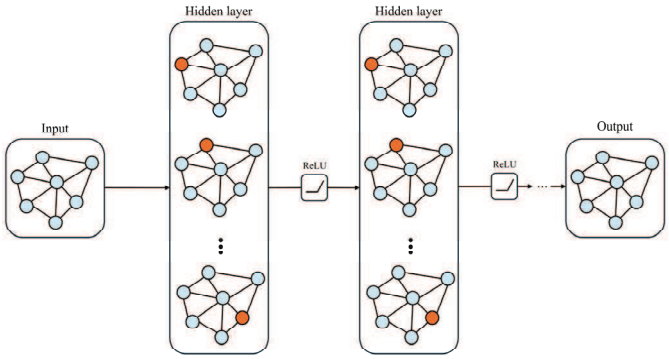


Fig. 3. Structure of a basic GCN.

$$H^{(l+1)} = \sigma\left(\hat{A}H^{(l)}W^{(l)}\right) \tag{2}$$

$$\hat{A} = D^{-\frac{1}{2}}(A + I)D^{-\frac{1}{2}} \tag{3}$$

where $H^{(l)} \in \mathbb{R}^{N \times F_l}$ is the node feature matrix at the lth layer, with N representing the number of nodes and F_l denoting the feature dimension. $W^{(l)} \in \mathbb{R}^{F_l \times F_{l+1}}$ is the learnable weight matrix at the lth layer. The matrix $\hat{A} \in R^{N \times N}$ is the normalized adjacency matrix, where A is the adjacency matrix describing the connectivity between nodes ($A_{ij} = 1$ if nodes i and j are connected, otherwise $A_{ij} = 0$). The identity matrix I is included to preserve as self-connections, and D is the diagonal degree matrix, where $D_{ii} = \sum_j A_{ij}$ is the degree of node i. The function σ is a nonlinear activation function, such as the ReLU function. The updated feature matrix $H^{(l+1)}$ encodes the refined node representations.

By stacking multiple graph convolutional layers, the GCN gradually learns higher-order neighborhood information, enabling nodes to integrate contextual information from distant neighbors. The three-layer GCN structure is given by

$$Z = \mathrm{softmax}\left(\hat{A}\mathrm{Relu}\left(\hat{A}\mathrm{Relu}\left(\hat{A}XW^{(0)}\right)W^{(1)}\right)W^{(2)}\right) \tag{4}$$

where $X = H^{(0)}$ denotes the initial feature matrix of LCFs. The softmax function is a normalized exponential function that converts raw feature scores into probability distributions. The weight matrices $W^{(0)} \in \mathbb{R}^{N \times H}$, $W^{(1)} \in \mathbb{R}^{H \times H}$, and $W^{(2)} \in \mathbb{R}^{H \times F}$ correspond to the input to hidden, hidden to hidden, and hidden to output transformations, respectively.

2) IED Module: GCNs face two major challenges when applied to remote sensing data for LSA purposes. On the one hand, pixel-based methods primarily capture local spatial dependencies, focusing on individual pixels and their immediate neighbors. However, this approach may fail to account for broader regional patterns and geomorphological structures that are crucial for accurately delineating landslide-prone areas. Consequently, the model may struggle to generalize effectively to diverse terrains with complex geological conditions. On the other hand, while superpixel-based methods improve spatial coherence by grouping similar regions, they inherently assume uniformity within each superpixel. This assumption may overlook fine-scale variations in topography, lithology,

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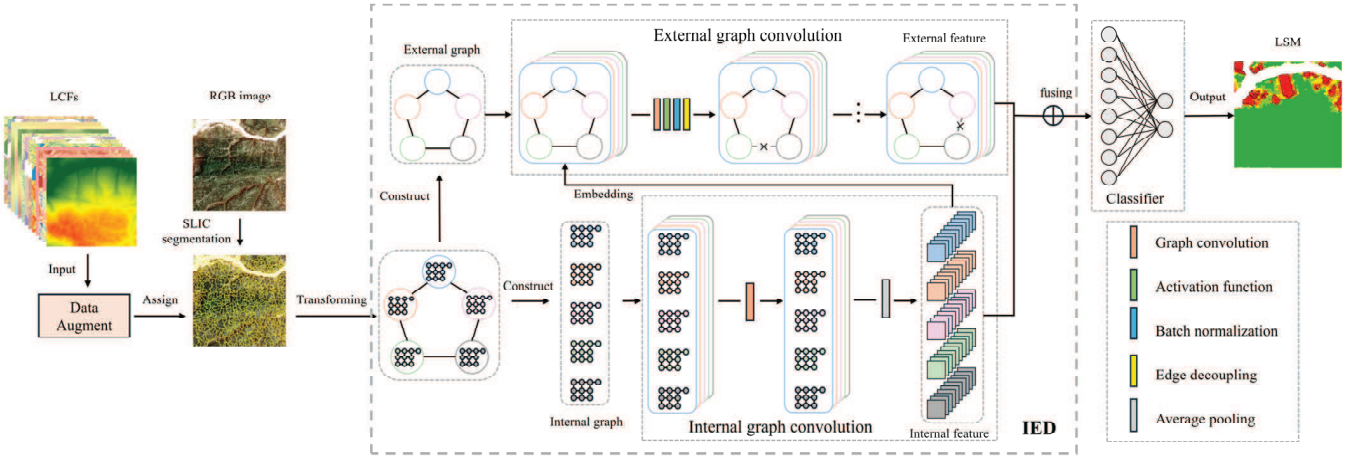


Fig. 4. Overall architecture of IED-GCN (the colors of the internal and external features are used solely for visual distinction of different superpixels).

and hydrological conditions within the same region, leading to potential information loss and reduced accuracy in feature representation.

To address these limitations, the proposed IED-GCN introduces significant advantages. By decomposing graph convolution into IGC and EGC, the model balances local feature extraction and global spatial context while preserving the internal heterogeneity of superpixels. IGC refines feature representation within superpixels, capturing fine-scale geological and topographical variations relevant to LSA. Simultaneously, EGC models spatial dependencies between superpixels, ensuring that broader geomorphological patterns and environmental interactions are effectively integrated into LSA. In addition, to mitigate the risk of overfitting caused by strong graph connectivity, an edge decoupling mechanism is incorporated into EGC. By randomly removing a certain percentage of edges, this approach reduces structural redundancy and prevents excessive dependence on densely connected nodes, allowing the model to focus on essential spatial relationships. This not only enhances the model's generalization ability but also improves its capacity to distinguish landslide-prone regions in complex terrains.

In the proposed model, the input LCF features can be expressed as $X \in \mathbb{R}^{H\times W\times C}$, where H and W denote the height and width of the image, respectively, and C is the number of LCFs. The image is segmented into S superpixels $A = \{A_1, \ldots, A_s\}$ using the simple linear iterative clustering (SLIC) algorithm [46], where each superpixel $A_s = \{x_{1s}, \ldots, x_{\text{ns}}\}$ contains ns pixels. During data preprocessing, each A_s is transformed into an internal graph $G_s = \{A_s, E_s\}$, and subsequently, an external graph $G_{\text{ex}} = \{G_{\text{in}}, E_{\text{ex}}\}$ is constructed. Here, G_{in} represents the set of nodes in the external graph and corresponds to the set of internal graphs. In the internal graph, the feature matrix of a superpixel A_s is denoted as X_{ins}, with an adjacency matrix A_{ins}. The output feature is computed as $h_s = f^{\text{in}}_{*_G}(X_{\text{ins}}, A_{\text{ins}})$. After processing all internal graphs, an internal feature matrix, $H_{\text{in}} = [h_1, \ldots, h_s]^{\text{T}}$, is obtained. In the external graph, since each node of G_{ex} represents an internal graph, the node feature matrix of the external graph is defined as $X_{\text{ex}} = H_{\text{in}}$, with an adjacency matrix A_{ex}. The output feature matrix of the external graph is expressed as $H_{\text{ex}} = f^{\text{ex}}_{*_G}(H_{\text{in}}, A_{\text{ex}})$. Finally, the internal and external feature matrices are fused. Sections III-B.1 and III-B.2 detail the construction and convolution of internal and external graphs, respectively. The overall architecture of our IED-GCN model is shown in Fig. 4.

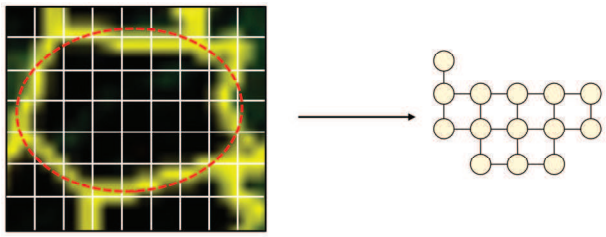


Fig. 5. Internal graph representation: each superpixel corresponds to an internal graph, in which each pixel is connected to its four neighboring pixels.

3) Internal Graph Construction and Convolution: As shown in Fig. 5, a superpixel A_s is converted into an internal graph G_s, where each pixel is connected to its four surrounding pixels (up, down, left, and right). Each pixel is represented as a node within the internal graph, with its corresponding node feature matrix denoted as $X_{\text{ins}} = [x_{1s}, \ldots, x_{\text{ns}}]^{\text{T}}$, containing a total of ns pixels. The edge set, $E_s = \{(x_{is}, x_{js})\}$, defines the connectivity among pixels within the superpixel A_s. Consequently, the set of internal graphs is represented as $G_{\text{in}} = \{G_1, \ldots, G_s\}$, on which IGC is performed. The superpixel IGC is formulated as follows:

$$f^{\text{in}}_{*_G}(X_{\text{ins}}, A_{\text{ins}}) = \text{pool}_{\text{mean}}\left(\tilde{X}_{\text{ins}}\right) \tag{5}$$

$$\tilde{X}_{\text{ins}} = \left(\tilde{D}_S^{-\frac{1}{2}}\tilde{A}_{\text{ins}}\tilde{D}_S^{-\frac{1}{2}}\right)^K X_{\text{ins}} W_{\text{in}} \tag{6}$$

where X_{ins} is the feature matrix, A_{ins} is the adjacency matrix, $\tilde{D}_S$ is the node degree matrix, and $\tilde{A}_{\text{ins}} = A_{\text{ins}} + I$. The parameter K represents the propagation depth of GCN node information at each layer, while W_{in} denotes the weight matrix governing pixel relationships between the internal graphs. At

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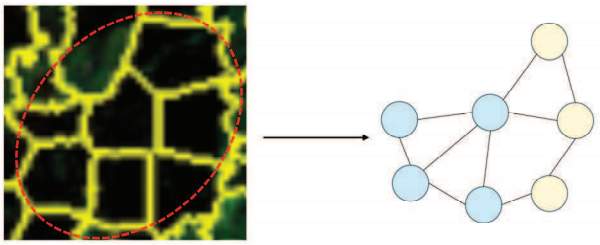


Fig. 6. External graph. A superpixel block represents a node.

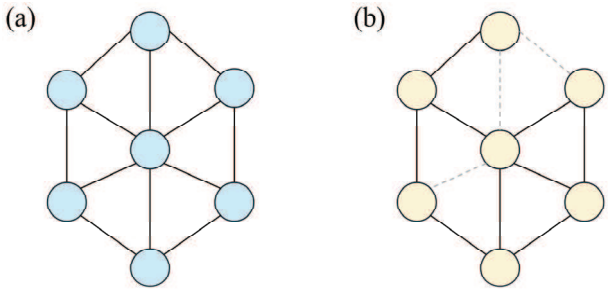


Fig. 7. Edge decoupling schematic, where (a) indicates that no edge is discarded and (b) indicates that part of the edges are discarded, the solid line indicates the connection, and the dashed line indicates the unconnected nodes.

the same time, to prevent overfitting and gradient explosion during training, L1 regularization and gradient clipping are incorporated into the loss function. L1 regularization encourages sparsity by penalizing the sum of absolute values of its elements. Gradient clipping is applied to constrain the maximum gradient magnitude during backpropagation. These techniques are formulated as follows:

$$\mathcal{L}_{\mathrm{L1}} = \lambda \sum_{i,j} \left| W_{ij} \right| \tag{7}$$

$$g = \begin{cases} g, & \text{if } \|g\| \le c \\ \dfrac{c}{\|g\|} g, & \text{if } \|g\| > c \end{cases} \tag{8}$$

where λ is the regularization coefficient, which controls the strength of the L1 penalty, g is the gradient, and c is the predefined clipping threshold. The final loss function L can be expressed as

$$\mathrm{L} = \mathcal{L}_{\mathrm{loss}} + \mathcal{L}_{\mathrm{L1}} + \mathcal{L}_{\mathrm{clip}} \tag{9}$$

where $\mathcal{L}_{\mathrm{loss}}$ is the original cross-entropy loss function, $\mathcal{L}_{\mathrm{L1}}$ accounts for L1 regularization, and $\mathcal{L}_{\mathrm{clip}}$ represents the gradient clipping term. These enhancements improve model sparsity, enabling the GCN to focus more effectively on local spectral correlations within the data.

4) External Graph Construction and Convolution: Superpixel segmentation disrupts the original pixel connectivity, making it insufficient to perform only IGC. To address this, an external graph is constructed in addition to the internal graph. As shown in Fig. 6, the external graph is formed by treating each superpixel as a node.

Formally, the external graph is denoted as $G_{\mathrm{ex}} = \{G_{\mathrm{in}}, E_{\mathrm{ex}}\}$, where the node set $G_{\mathrm{in}} = \{G_1, \ldots, G_s\}$ comprises the internal graphs and the edge set $E_{\mathrm{ex}} = \{(A_i, A_j)\}$ represents the connections between internal graphs. The EGC enables feature propagation between superpixels and is defined as

$$H_{\mathrm{ex}}^{l+1} = \mathrm{ReLU}\left(\tilde{D}_{\mathrm{ex}}^{-\frac{1}{2}} \tilde{A}_{\mathrm{ex}} \tilde{D}_{\mathrm{ex}}^{-\frac{1}{2}}\right) H_{\mathrm{ex}}^{l} W_{\mathrm{ex}}^{l} \tag{10}$$

where, ReLU is the activation function, $\tilde{D}_{\mathrm{ex}}$ is the node degree matrix of the external graph, and $\tilde{A}_{\mathrm{ex}} = A_{\mathrm{ex}} + I$, where A_{ex} is the adjacency matrix. H_{ex}^{l} represents the feature matrix at the lth layer, with the initial input $H_{\mathrm{ex}}^{0} = H_{\mathrm{in}} = [h_{\mathrm{in}1}, \ldots, h_{\mathrm{ins}}]^{\mathrm{T}}$. W_{ex}^{l} is the weight matrix of the lth layer. However, an excessive depth in EGC may increase the receptive field of nodes, leading to over-smoothing and a reduction in model performance. To mitigate this, the number of layers is set to four, and an edge decoupling mechanism is employed to enhance generalization.

Edge decoupling randomly removes a subset of edges during training, thereby avoiding overfitting and improving robustness (see Fig. 7).

The strategy of discarding edges allows GCN to train on an incomplete graph structure for each round of training, which helps to increase the robustness of the model to structural noise and makes the model not dependent on some specific edges. This involves masking the adjacency matrix with a randomly generated binary matrix, $M \in \mathbb{R}^{N \times N}$, where each element M_{ij} is sampled from a Bernoulli distribution

$$M_{ij} \sim \mathrm{Bernoulli}(1 - p) \tag{11}$$

where p is the edge discarding rate, set to 0.1 in our experiment. The modified adjacency matrix is then computed as

$$A' = A \odot M \tag{12}$$

where A' is the modified adjacency matrix, A is the adjacency matrix, and $\odot$ represents element-wise multiplication.

The edge decoupling process effectively removes edges in each forward propagation, reducing dependence on specific graph topologies and improving the model's ability to generalize.

The architecture of our IED-GCN is shown in Fig. 4. First, the division of superpixels is carried out. Each superpixel is constructed into an internal graph, and all superpixels together form an external graph. Thus, the IE structure is successfully constructed. Then, the IED-GCN model is constructed to extract features by layers. Internal features are obtained through the IGC, and these internal features are used as input features to the EGC, thus obtaining external features. Finally, the internal features H_{in} and external features H_{ex} are fused to get H_{all}, and the final probability value is obtained by a multilayer perceptron (MLP) module

$$H_{\mathrm{all}} = H_{\mathrm{in}} + H_{\mathrm{ex}} \tag{13}$$

$$\mathrm{Output} = \mathrm{MLP}\left(H_{\mathrm{all}}\right). \tag{14}$$

It is worth noting that H_{all} covers the information of superpixels from small scale to large scale and from internal to external, instead of simple multihop neighbor information used in most existing GCN-based methods.

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C. Model EMs

To comprehensively evaluate the performance of the constructed LSA model, a series of widely used evaluation metrics (EMs) were adopted in this study. These EMs include the area under the receiver operating characteristic curve (AUC), overall accuracy (OA), Precision, Recall, F1 score (F1), Matthews correlation coefficient (MCC), and Kappa coefficient (Kappa). Each of these EMs provides a distinct perspective on the model's classification capability and the regression error [47]. AUC evaluates the classification performance of the model, with values closer to 1 indicating superior performance.

1) OA reflects the proportion of the total samples that the model predicts correctly.
2) Precision quantifies the proportion of correctly identified positive samples among those predicted as positive.
3) Recall represents the proportion of actual positive samples that are correctly classified by the model.
4) F1 is a harmonic average mean of Precision and Recall, balancing these two aspects of classification performance.
5) MCC evaluates the overall quality of the classification results, and is particularly suitable for unbalanced datasets.
6) Kappa assesses the agreement between predicted and actual results, accounting for chance agreement.

The mathematical definitions of these EMs are as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{15}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{16}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{17}$$

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{18}$$

$$\mathrm{MCC} = \frac{\mathrm{TP} \times \mathrm{TN} - \mathrm{FP} \times \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP}) \times (\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}} \tag{19}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - P_e}{1 - P_e}. \tag{20}$$

Among these, true positive (TP) indicates the number of samples correctly predicted by the model as landslides. True negative (TN) represents the number of samples that are correctly predicted by the model as nonlandslide. False positive (FP) denotes the number of samples incorrectly predicted as landslides by the model. False negative (FN) indicates the number of samples that are incorrectly predicted as nonlandslide by the model. P_e represents the expected accuracy under random classification. A value close to 1 for these EMs indicates superior model performance.

For LSA, the natural breakpoint method is usually used to categorize landslide susceptibility into five distinct levels: very high (VeH), high (Hi), moderate (Mo), low (Lo), and very low (VeL). To measure the consistency and rationality of landslide distribution and landslide susceptibility zonation, landslide density (LD) was used in this study to quantitatively describe the relationship between the predicted and the actual landslide areas. The formula is as follows:

$$\mathrm{LD} = \mathrm{PL}/\mathrm{PC} \tag{21}$$

where PC represents the proportion of the area of each prone zone relative to the entire study area, and PL denotes the proportion of the landslide area within the prone zone in relation to the whole study area. LD reflects the spatial relationship between landslides and landslide-prone areas. Generally, LD decreases as the landslide susceptibility grade decreases [48], [49]. A higher LD value indicates a more concentrated occurrence of landslides [50].

TABLE III
COMPUTER HARDWARE INFORMATION

Item	Detailed parameter
CPU	Intel Xeon (R) Silver 4210R CPU@2.4GHZ×40
GPU	GeForce RTX 3090/PCLe/SSE2
Memory	128GB
Hard disk	8TB

D. Experimental Environment and Hyperparameter Settings

Table III shows the hardware configuration used in the experiments conducted in this study. All experiments were executed on a computer equipped with an NVIDIA GeForce RTX 3090 GPU, running the Linux-Ubuntu 18.04.5 LTS operating system. The primary software environment includes Python 3.9, with the PyTorch 1.10.1 framework, complemented by commonly used scientific libraries, such as NumPy and Matplotlib, for auxiliary tasks.

The performance of a model, in terms of both accuracy and computational efficiency, is influenced not only by its architectural design but also by the careful tuning of hyperparameters [51]. The model architecture utilized 128 units in the hidden layers and was trained for 150 epochs, with one internal graph convolutional layer and four external graph convolutional layers. Then, SLIC [46] segmentation was applied to the input data for efficient image segmentation. The Adam optimizer was used with a learning rate of 10^{-3}, and the target function was regularized with a weight decay factor of 10^{-3} to mitigate the risk of overfitting. To address potential issues with exploding gradients, gradient clipping was implemented during the training process. Furthermore, an edge discard rate of 0.1 was applied during edge decoupling to enhance the model's robustness. The batch size was set to 64 to facilitate efficient computation and ensure stable convergence during training. An important hyperparameter in the model, discussed in Section III-B, is the number of superpixels S, which significantly affects segmentation quality and performance. A small value of S reduces computational efficiency, while a large S may yield coarse model results. To better capture the multiscale characteristics of landslides, the number of pixels

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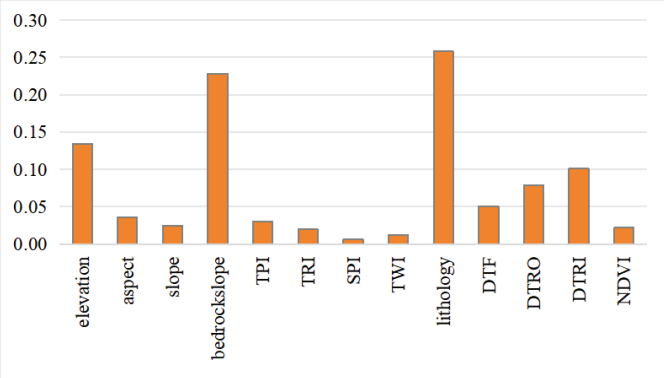


Fig. 8. Results of importance evaluation of LCFs.

per superpixel, denoted as n_p, was introduced as an indirect determinant of S, as given by the following equation:

$$S = \frac{H \times W}{n_p} \tag{22}$$

where H and W represent the height and width of the image, respectively. Due to the variable sizes of landslides, we tested the EMs under n_p = 10, 70, and 145. The three n_p values represent the minimum landslide grid count, the median grid count of all landslides, and the average grid count of all landslides. A detailed comparative analysis of these settings will be presented in Section V-B. The experimental results show that n_p = 10 performs best on all EMs. Therefore, we adopt n_p = 10 as the default parameter setting for this study.

IV. Results and Analysis

A. Analysis of LCFs

To evaluate the significance of the 13 adopted LCFs, we conducted a multicollinearity test and importance ranking based on VIF and Gini coefficients. The results are shown in Table IV and Fig. 8, respectively. The VIF analysis confirmed that all 13 LCFs exhibited VIF values below 10, indicating the absence of severe multicollinearity issues within the dataset. This result ensures that all LCFs can be retained for subsequent analysis [52], [53].

According to the Gini coefficient ranking, lithology, bedrockslope, and elevation emerged as the most influential factors. These findings are expected, as variations in terrain features directly affect the occurrence of landslides, reinforcing the dominant role of these three factors in LSA. Conversely, the lower importance of SPI and TWI suggests that their impact on landslide susceptibility is comparatively weaker. However, these factors may still contribute auxiliary information, particularly in complex geographical environments. SPI aids in identifying topographic features associated with landslide-prone zones, such as valley floors and concave landforms, while TWI provides insights into water accumulation and soil saturation. High TWI values usually indicate excessive moisture retention, increasing the likelihood of slope instability. Given these considerations, all 13 LCFs were incorporated into the LSA framework to ensure a comprehensive assessment.

TABLE IV
VIF Analysis Results for LCFs

LCFs	Multicollinearity statistics	
	VIF	TOL
Elevation	2.8348	0.3528
Aspect	1.1458	0.8728
Slope	7.8925	0.1267
Bedrockslope	3.8196	0.2618
TPI	4.5566	0.2195
TRI	3.1791	0.3146
SPI	1.2112	0.8256
TWI	3.5529	0.2815
lithology	3.7056	0.2699
DTF	1.1268	0.8874
DTRO	2.6009	0.3845
DTRI	1.6689	0.5992
NDVI	1.6925	0.5908

B. Comparison of LSA Results

To evaluate the effectiveness of the proposed IED-GCN, we compared it with CNN, ResNet, DenseNet, and miniGCN. CNN-based models were selected due to their strong capability in processing raster data and their feature extraction mechanism by aggregating local neighborhood information, which is similar to that of GCNs. miniGCN was chosen because it uses a patch-based mini-batch input approach that is structurally comparable to the superpixel segmentation method employed in IED-GCN. The natural breakpoint method was employed to categorize the landslide susceptibility into five levels: VeL, Lo, Mo, Hi, and VeH [54], [55]. The spatial distribution of susceptibility levels predicted by each model is shown in Fig. 9. The results indicate that most of the study area falls within the VeL and Lo susceptibility zones, whereas the Hi and VeH zones are concentrated in specific regions corresponding to historical landslide occurrences. Notably, landslide susceptibility decreases with increasing distance from water bodies. However, certain nonhistorical landslide-prone areas adjacent to the water system were predicted as medium and high susceptibility zones, suggesting potential risk areas for future landslides. Among the evaluated models, IED-GCN stands out for its ability to align Hi and VeH zones more precisely with historical landslide data, particularly in capturing fine-grained details and boundary accuracy.

To further quantify model performance, we evaluated the PC, PL, and LD values for each model, as summarized in Table V. Notably, IED-GCN exhibits superior performance in terms of spatial accuracy and predictive capability. It identified a historically high percentage of landslides within the Hi and VeH zones of 97.81%, and, while this is slightly lower than the CNN model's 99.09%, it minimizes overprediction of landslide-prone areas, with the Hi and VeH zones

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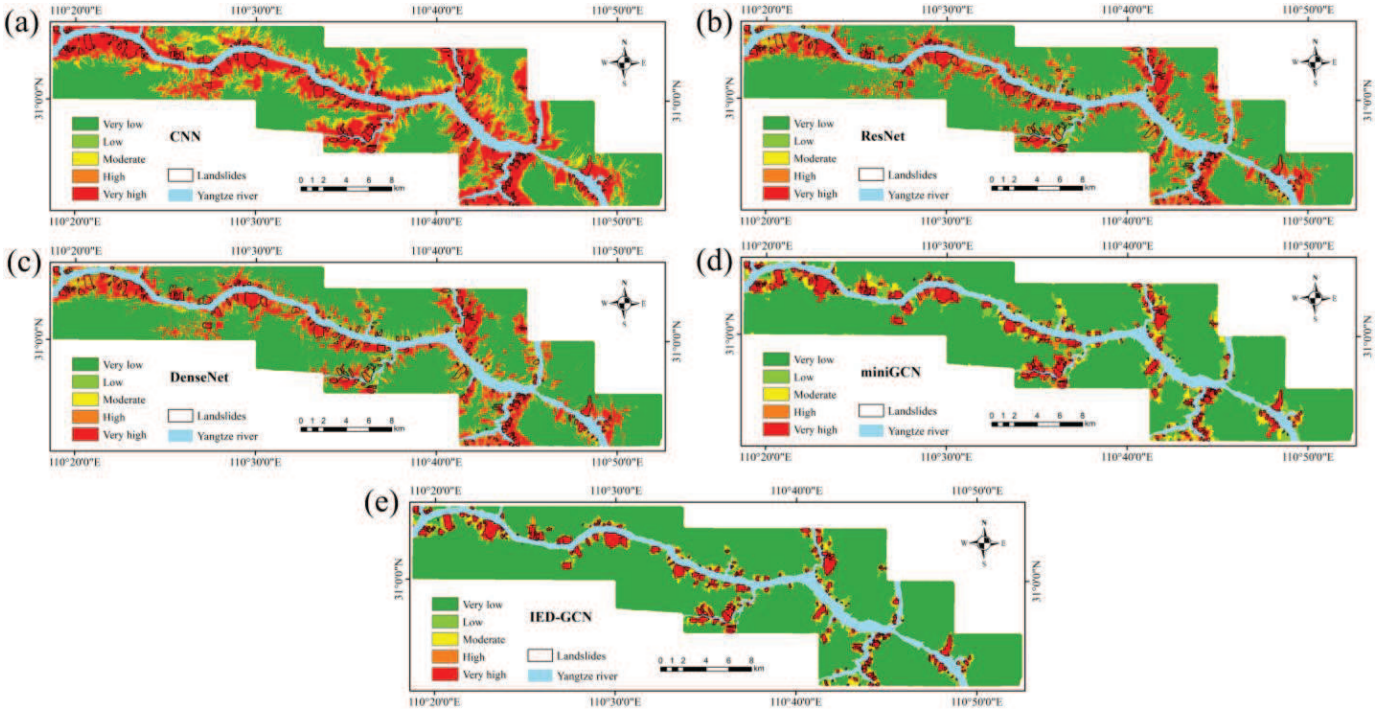


Fig. 9. LSA results obtained from five models. (a) CNN, (b) ResNet, (c) DenseNet, (d) miniGCN, and (e) IED-GCN.

accounting for only 10.25%. LD metric combines PC and PL to provide a quantitative measure of how well the model distributes landslide predictions across susceptibility categories. As shown in Table V, LD values for all models increased as susceptibility levels increased, with the VeH region consistently displaying the highest LD values. The IED-GCN model consistently exhibited the highest LD values within the VeH region, reinforcing its strong correlation with actual landslide distributions [56]. Furthermore, IED-GCN demonstrated the largest variation in LD values across different susceptibility levels, highlighting its effectiveness in distinguishing stable from unstable regions.

In summary, the obtained LSA results showed that IED-GCN exhibited superior spatial consistency and quantitative alignment with historical landslides. Its ability to accurately predict high-susceptibility zones while minimizing FPs highlights its potential as a reliable tool for LSA.

C. Comparative Evaluation of Model Performance

To ensure the objectivity and reliability of the model evaluation, the dataset was divided into a training set and a test set, with the training set used for model training and hyperparameter tuning, while the test set was reserved for performance evaluation. To comprehensively evaluate the performance of the model, a variety of EMs were used, including AUC, OA, Precision, Recall, F1, MCC, and Kappa. The quantitative results of these EMs are shown in Table VI, with the optimal values for each EM highlighted in bold to facilitate a direct comparison of model performance.

Among all models, the proposed IED-GCN model demonstrated the best performance, achieving an AUC of 0.9838, which is significantly higher than other comparative models. The AUC value reflects the model's classification capability across different threshold settings, and a high AUC value confirms the model's accuracy and robustness in distinguishing landslide-prone regions from nonlandslide-prone regions. This result underscores the superiority of the IED-GCN model in the task of LSA.

Beyond AUC, the IED-GCN also exhibited superior performance across other EMs. The OA reached 0.9838, indicating the model's ability to correctly classify the vast majority of samples, thereby achieving the highest OA. The Precision value of 0.9772 suggests that, among the samples predicted as landslide-prone, the proportion of actual landslide occurrences was high, minimizing the FP rate. This is a crucial attribute in reducing unnecessary hazard alerts [12]. Furthermore, the F1 score of 0.9839 validates the model's ability to maintain a balance between Precision and Recall, highlighting its classification stability and reliability. When evaluating the model's ability to identify landslide-prone areas, the Recall value of 0.9907 demonstrates its effectiveness in detecting most landslide occurrences, even in cases where landslide samples are relatively scarce. This high Recall performance is particularly crucial for disaster risk management, ensuring that key hazard zones are not overlooked, which aligns with findings from prior landslide prediction studies.

In addition, the IED-GCN model achieved MCC and Kappa values of 0.9676 and 0.9675, respectively. The MCC metric comprehensively considers TPs, TNs, FPs, and FNs, providing a balanced evaluation of the model's classification performance, especially in cases of imbalanced class distributions. Meanwhile, the Kappa coefficient measures the consistency between model predictions and actual obser-

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TABLE V
LD VALUE FOR ALL METHODS

Models	LSA partition	PC	PL	LD
CNN	VeL	0.4517	0.0013	0.0029
	Lo	0.0891	0.0023	0.0257
	Mo	0.0760	0.0054	0.0713
	Hi	0.0942	0.0460	0.4890
	VeH	0.2890	0.9449	3.2698
ResNet	VeL	0.6462	0.0398	0.0616
	Lo	0.0595	0.0487	0.8177
	Mo	0.0533	0.0783	1.4681
	Hi	0.0706	0.1701	2.4097
	VeH	0.1704	0.6631	3.8912
DenseNet	VeL	0.5885	0.0179	0.0304
	Lo	0.0543	0.0171	0.3148
	Mo	0.0476	0.0336	0.7062
	Hi	0.0842	0.1570	1.8632
	VeH	0.2253	0.7745	3.4373
miniGCN	VeL	0.7757	0.0032	0.0041
	Lo	0.0563	0.0142	0.2530
	Mo	0.0376	0.0313	0.8329
	Hi	0.0419	0.1186	2.8330
	VeH	0.0885	0.8326	9.4034
IED-GCN	VeL	0.8277	0.0008	0.0010
	Lo	0.0396	0.0036	0.0921
	Mo	0.0302	0.0174	0.5773
	Hi	0.0261	0.0677	2.5969
	VeH	0.0764	0.9104	11.9114

vations beyond random chance, with a high Kappa value further reinforcing the model's stability and reliability [57]. In conclusion, the IED-GCN model demonstrated outstanding performance in LSA tasks, exhibiting high accuracy, robustness, and generalization capability. Its superior ability to identify landslide-prone regions with minimal false alarms makes it a highly reliable tool for landslide disaster prevention and management.

D. Ablation Experiments

To verify the validity of the ICG, EGC, and edge decoupling modules in the proposed model, we conducted ablation experiments comparing models traditional baseline GCN, utilizing only IGC alone, EGC alone, and both IGC and EGC (IE-GCN), and the proposed IED-GCN model with IGC, EGC, and edge decoupling. In all experimental settings, training procedures and hyperparameters remained consistent, with module selection being the only variable. The results, summarized in Table VII, demonstrate the significant impact of different module combinations on the model performance.

When using only IGC, the model performed feature aggregation within superpixels, effectively capturing local spatial structures. This enhanced regional feature representation improves the model's ability to characterize geological disaster-prone areas. However, due to the lack of global information interaction between superpixels, overall predictive accuracy declined, especially in complex terrain where cross-region influences are critical. For example, the AUC value dropped to 0.8651 and the F1 score was 0.8699, indicating that relying solely on internal superpixel features is insufficient for comprehensive LSA.

With only EGC, the model mainly captured spatial correlations between regions through feature propagation across superpixels. This compensated for IGC's limitations by improving global perception. However, due to the lack of fine-grained modeling of internal superpixel structures leads to oversmoothing, reducing the model's ability to delineate geological hazard boundaries. The AUC value increased to 0.8868, and the F1 score reached 0.8934, which is higher than IGC alone, but still inferior to the IED-GCN model.

Since the IGC and EGC modules function synergistically within the model, using either IGC or EGC alone results in limited capability to capture spatial information, leading to overall performance that is lower than the traditional baseline GCN. Nevertheless, in the IE-GCN experiment (using both IGC and EGC but without edge decoupling), the model leveraged both local and global spatial information, significantly enhancing predictive performance compared to using IGC or EGC alone. Moreover, compared to the baseline GCN without any modular enhancement, IE-GCN achieved superior performance across multiple EMs, further validating the effectiveness of the combined use of IGC and EGC. However, due to the retention of all superpixel connections, the model suffered from redundant edge information, resulting in feature mixing and reduced generalization. The results showed an AUC of 0.9810, with a slight decline in Recall (0.9878) and MCC (0.9621), indicating that excessive connectivity introduced noise.

Overall, these ablation experiments confirm that IGC and EGC jointly determine the model's baseline performance—IGC is responsible for local feature extraction, while EGC facilitates global information propagation. Their combined use in IE-GCN significantly enhances predictive accuracy. However, the rigid fixed connectivity of superpixels introduces redundancy, reducing model robustness. By integrating an edge decoupling mechanism, IED-GCN effectively captures informative spatial features while mitigating the influence of irrelevant connections, thereby improving the accuracy and reliability of landslide susceptibility prediction.

V. DISCUSSIONS

A. Analysis of Model Stability and Landslide Spatial Differences

Model stability is a critical factor in ensuring the reliability and consistency of LSA. A stable model should exhibit minimal performance variation across multiple training iterations, demonstrating robustness to fluctuations in input data. To systematically evaluate the stability of the proposed model, we

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TABLE VI
MODEL EMS OF EACH MODEL (BOLDED VALUE MEANS THE BEST VALUE IN EACH COLUMN)

Models	OA	Precision	Recall	F1	MCC	Kappa	AUC
CNN	0.9507	0.9209	0.9861	0.9524	0.9037	0.9014	0.9442
ResNet	0.9654	0.9511	0.9811	0.9659	0.9312	0.9307	0.9627
DenseNet	0.9595	0.9395	0.9823	0.9604	0.9200	0.9190	0.9588
miniGCN	0.9720	0.9584	0.9869	0.9724	0.9445	0.9440	0.9753
IED-GCN	**0.9838**	**0.9772**	**0.9907**	**0.9839**	**0.9676**	**0.9675**	**0.9838**

TABLE VII
ABLATION RESULTS (BOLDED VALUE MEANS THE BEST VALUE IN EACH COLUMN)

Models	IGC	EGC	Edge decoupling	OA	Precision	Recall	F1	MCC	Kappa	AUC
Baseline model (GCN)				0.9634	0.9499	0.9785	0.9640	0.9273	0.9268	0.9640
IGCN	√			0.8651	0.8403	0.9017	0.8699	0.7322	0.7303	0.8654
EGCN		√		0.8868	0.8442	0.9486	0.8934	0.7795	0.7736	0.8868
IE-GCN	√	√		0.9810	0.9746	0.9878	0.9812	0.9621	0.9621	0.9810
IED-GCN	√	√	√	**0.9838**	**0.9772**	**0.9907**	**0.9839**	**0.9676**	**0.9675**	**0.9838**

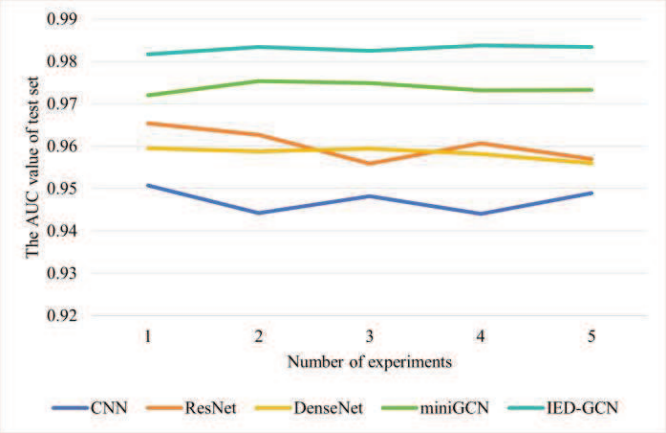


Fig. 10. Stability estimates of all models.

randomly select nonlandslide samples five times and combine them with the original landslide samples to generate five independent datasets, while maintaining a fixed training-to-test sample ratio of 7:3. Each dataset was used to train and validate five different models, generating five sets of predictions per model. To visualize performance variation across different training instances, the AUC values for each experiment were computed and plotted in Fig. 10. Models exhibiting high stability should show minimal fluctuations in AUC values across different runs, indicating resilience to dataset variability.

As can be seen from the results, the IED-GCN model consistently achieved the highest AUC values across all five experimental runs. Moreover, IED-GCN demonstrated lower fluctuation than CNN and ResNet models, indicating superior stability and reduced sensitivity to data perturbations.

To further evaluate the spatial consistency of model predictions, we randomly selected 30 000 grid cells from the study area to assess prediction consistency across different models. For each grid cell, the average susceptibility probability derived from the five datasets (x-axis) was compared with the highest susceptibility probability predicted by the best-performing model (y-axis), as shown in Fig. 11. To quantify the degree of consistency, the coefficient of determination (R^2) was calculated, where a higher R^2 value indicates a stronger correlation between the average and the best-case susceptibility values, reflecting greater model stability and lower sensitivity to input variations.

The results indicate that all models exhibit low uncertainty in regions classified as VeH and VeL susceptibility, while greater uncertainty is observed in the Mo susceptibility region. Among the evaluated models, IED-GCN produced the most concentrated scatterplot along the fitting curve and achieved the highest R^2 value of 0.9545, further confirming its superior stability, spatial prediction consistency, and resilience to input variations.

Beyond stability, a model's ability to capture landslide spatial differences is critical for accurately delineating high-risk areas. To assess this capability, we extracted the predicted susceptibility values for historical landslide regions across all five models and analyzed their distribution characteristics and central tendency (see Fig. 12). In the violin plot, the wider section indicates higher pixel density within that range, while longer tails suggest outliers in susceptibility predictions. The 25%–75% interquartile range (dotted line) represents the core distribution of predicted values, serving as an indicator of prediction consistency and stability. In addition, the median line (solid line) highlights the central tendency, providing a statistical summary of each model's susceptibility predictions.

As can be seen from Fig. 12, IED-GCN exhibits the highest mean susceptibility value (Mean) in historical landslide areas. Furthermore, its median line and 25%–75% interquartile range

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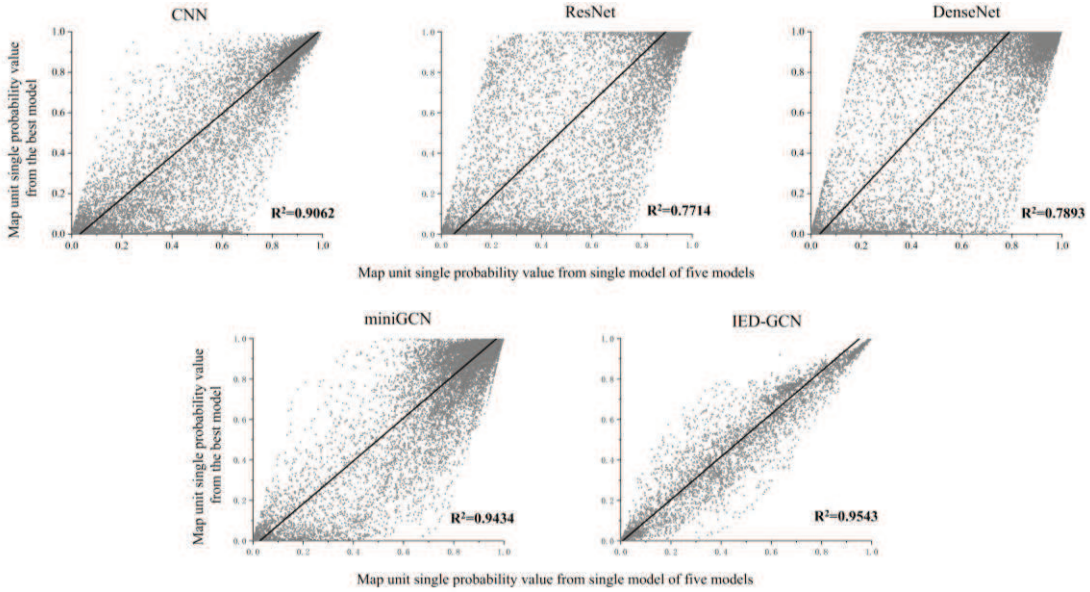


Fig. 11. Uncertainty of the results predicted by all models (x-axis is the map unit mean probability value from models of five datasets, y-axis is the map unit single probability value from the "best" model, and R^2 is the consistency between variance x and y).

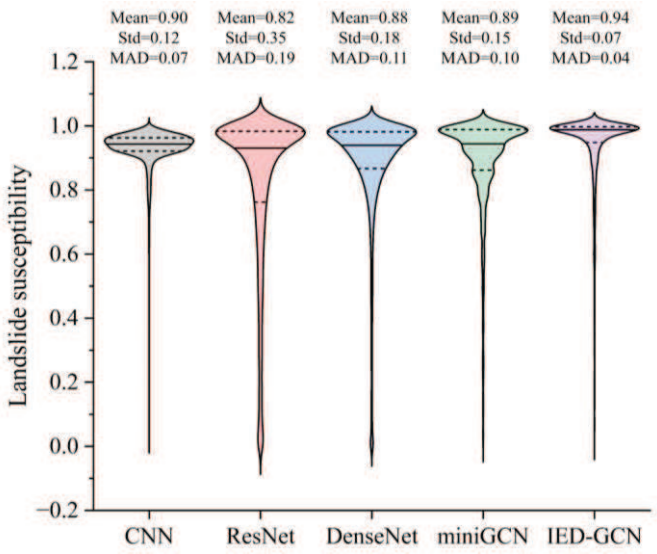


Fig. 12. Distribution of predicted susceptibility values in historical landslide regions.

are higher than those of the other models, indicating that its predictions are more concentrated in landslide-prone areas and that it more accurately identifies high-risk areas. The standard deviation (Std) and the mean absolute deviation (MAD) of IED-GCN are also the lowest among all models, suggesting greater stability in predicting landslide-prone regions and reduced sensitivity to input data fluctuations. In contrast, the susceptibility value distributions of ResNet, DenseNet, and miniGCN models are more dispersed, with higher variability, indicating greater uncertainty in their landslide susceptibility characterization.

Overall, IED-GCN outperforms the other models in both stability and fine-detail capturing ability. Its AUC values exhibit the least fluctuation across different runs, and it achieves the highest R^2 value, demonstrating superior robustness to input data variations. In addition, its susceptibility predictions in historical landslide regions are more concentrated, with the lowest Std and MAD, confirming its ability to provide stable and precise LSA. In contrast, CNN, ResNet, DenseNet, and miniGCN show more dispersed predictions with higher uncertainty. These findings highlight the reliability of IED-GCN for LSA.

B. Analysis of Different Parameters

The edge discard rate and superpixel block size n_p are two critical hyperparameters influencing the performance of the proposed IED-GCN model. To explore their effects on the model's predictive ability, comparative experiments were conducted by systematically adjusting these parameters and analyzing their effects on model stability and accuracy.

For the edge discard rate, experimental groups were evaluated at 0%, 10%, 20%, and 30% (see Table VIII). The results indicate that a 10% edge discard rate optimally reduces irrelevant information propagation, yielding the best performance across key Ems, such as AUC, F1, and MCC. However, when the edge discard rate exceeds 20%, model performance declines, especially in the aspect of Recall. This suggests that excessive edge pruning may lead to information loss, impairing the model's ability to effectively capture the internal spatial relationships within superpixel regions. Hence, a moderate edge discard rate is necessary to enhance model stability and adaptability in geological hazard prediction.

For superpixel block size (n_p), sizes of 10, 70, and 145 were analyzed (see Table IX). The results indicate that an n_p of 10 yields the best overall performance, balancing local details and global information. Smaller superpixel blocks preserve edge information and fine geological features, enhancing spatial resolution in complex terrain while ensuring effective

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TABLE VIII
PERFORMANCE OF IED-GCN WITH DIFFERENT EDGE DISCARD RATE (BOLDED VALUE MEANS THE BEST VALUE IN EACH COLUMN)

Edges discard rate	OA	Precision	Recall	F1	MCC	Kappa	AUC
0%	0.9810	0.9746	0.9878	0.9812	0.9621	0.9621	0.9810
10%	**0.9838**	**0.9772**	**0.9907**	**0.9839**	**0.9676**	**0.9675**	**0.9838**
20%	0.9807	0.9731	0.9889	0.9809	0.9616	0.9615	0.9806
30%	0.9801	0.9769	0.9835	0.9802	0.9602	0.9602	0.9802

TABLE IX
PERFORMANCE OF IED-GCN WITH DIFFERENT n_p (BOLDED VALUE MEANS THE BEST VALUE IN EACH COLUMN)

n_p	OA	Precision	Recall	F1	MCC	Kappa	AUC
10	**0.9838**	**0.9772**	**0.9907**	**0.9839**	**0.9676**	**0.9675**	**0.9838**
70	0.9353	0.9068	0.9703	0.9375	0.8727	0.8705	0.9353
145	0.8818	0.8382	0.9462	0.8889	0.7700	0.7635	0.8818

TABLE X
MODEL PERFORMANCE UNDER DIFFERENT SCALE DATA (BOLDED VALUE MEANS THE BEST VALUE IN EACH COLUMN)

Sample proportion	OA	Precision	Recall	F1	MCC	Kappa	AUC
1:1	0.9838	**0.9772**	**0.9907**	**0.9839**	**0.9676**	**0.9675**	**0.9838**
1:2	0.9797	0.9623	0.9773	0.9698	0.9545	0.9545	0.9791
1:4	0.9806	0.9441	0.9598	0.9519	0.9397	0.9397	0.9728
1:8	0.9838	0.9313	0.9223	0.9268	0.9177	0.9177	0.9569
1:16	**0.9879**	0.9107	0.8830	0.8966	0.8903	0.8902	0.9387

learning of topological structure. When n_p increases to 70, some local details are lost due to broader superpixel coverage. Further increasing n_p to 145 significantly degrades model performance, likely due to the reduced connections that hinder GCN's ability to retain essential spatial structure information during information transmission.

To sum up, an edge discard rate of 10% and a n_p of 10 were found to be optimal for LSA, ensuring model generalization while preserving critical geological features.

C. Effect of Sample Scale on Model Performance

Class imbalance is a widely recognized challenge in LSA, as it can severely impact the model's ability to detect minority-class events [58], [59]. In real-world scenarios, landslide samples are often far fewer than nonlandslide samples, resulting in imbalanced datasets that may bias model learning and compromise detection performance. To systematically evaluate this effect, comparative experiments were conducted with varying sample ratios (1:1, 1:2, 1:4, 1:8, and 1:16) to assess their impact on model performance (see Table X). The results reveal that model performance is significantly affected by sample imbalance. At a balanced 1:1 ratio, the model achieves the best overall performance, except for OA, which is slightly lower. This suggests that an equal sample distribution enables better feature learning for both classes.

As the proportion of landslide samples decreases, model performance declines, particularly in OA and Recall. Severe imbalance (e.g., 1:16) leads to a substantial drop in landslide detection capability, as the model becomes biased toward non-landslide predictions. This highlights the detrimental impact of extreme class imbalance on model effectiveness.

D. Limitations and Future Work

Despite its strong performance, the IED-GCN model has certain limitations. First, its effectiveness relies on the quality and completeness of input data. While the combination of superpixel partitioning and graph convolution effectively captures local and global spatial features, data scarcity or poor quality may hinder predictive accuracy. In remote areas, incomplete landslide inventories may lead to inaccurate susceptibility assessments. Chen et al. [60] proposed a novel data fusion technique for landslide early warning based on generalized evidence theory, which integrates rainfall detectors, GNSS, and deep displacement sensors to capture both external triggers and internal states of landslides. This method effectively reduces uncertainty in hazard prediction. Future studies could explore multimodal remote sensing data integration (such as Sentinel-1 radar and UAV hyperspectral data) to supplement geographic information with time-series observation, improving spatial and temporal coverage.

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In addition, geographic data sources, such as DEM and Landsat-8, have varying spatial and temporal resolutions. Although resampling and preprocessing mitigate inconsistencies, information loss or noise may still affect model accuracy. Furthermore, the reliance on the SLIC superpixel partitioning algorithm introduces potential bias due to manually set parameters (e.g., size and compactness). Given terrain variability, fixed parameter settings may lead to information loss features in some regions. For example, in complex mountainous areas, too large superpixels may mask microtopographic changes, while too small superpixels may introduce redundant noise. Bai et al. [61] developed an adaptive graph structure mining (AGSM) mechanism to reduce manual intervention by dynamically constructing non-Euclidean graph structures that better aggregate spatial information. Future work should focus on developing adaptive superpixel generation techniques or integrating superpixel partitioning with graph structure learning to dynamically optimize local region aggregation, reducing reliance on manual tuning.

Despite these limitations, IED-GCN offers a completely novel approach to LSA. Future studies can further refine superpixel partitioning methods, optimize graph convolution training strategies, and construct high-quality, high-resolution landslide datasets to further improve the generalization and practical applicability.

VI. Conclusion

In this article, we propose a new LSA framework, termed IED-GCN. By integrating the hierarchical feature extraction mechanisms of IGC and EGC, the proposed framework fully captures both local and global spatial dependencies with superpixels. Specifically, IGC focuses on preserving fine-grained details within individual superpixels, while EGC models broader spatial interactions between superpixels through a global graph structure. In addition, the edge decoupling module introduced in EGC significantly reduces structural noise by randomly discarding redundant edges, allowing the model to concentrate on extracting key spatial relationships.

To verify the effectiveness of IED-GCN, we conducted extensive experiments using the Three Gorges Reservoir area as the study site. Compared against that of CNN, ResNet, DenseNet, and miniGCN, the experimental results demonstrate that IED-GCN significantly outperforms competing models in LSA within the Three Gorges Reservoir area. Notably, IED-GCN exhibits superior predictive precision, accurately delineating areas of high landslide susceptibility that closely align with historical landslide distributions. This highlights the model's robustness and practical applicability in spatial dependence modeling. In summary, the proposed IED-GCN framework effectively captures both local and global spatial features, enabling more accurate and reliable LSA. Moreover, its adaptability to complex spatial data environments makes it particularly well-suited for geological disaster prediction tasks, offering valuable insights for risk management and mitigation strategies.

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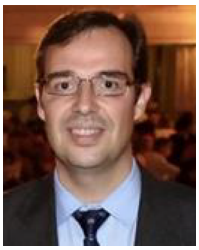
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